



Graphic Era
HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Even) Semester Regular/Back Examination May 2026

Roll no. 2294038.....

Name of the Program and semester: B. Tech., VIII

Name of the Course: Soft Computing

Course Code: TCS-821

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

1. a. Explain the structure and working of a Biological Neural Network. Compare it with Artificial Neural Networks highlighting key differences. (CO1)

b. Describe the building blocks of ANN and explain important terminologies such as architecture, weights, and activation functions with suitable examples. (CO1)

c. Explain the following learning rules with mathematical formulation and applications:

- (i) McCulloch-Pitts Neuron Model
- (ii) Hebbian Learning Rule
- (iii) Perceptron Learning Rule
- (iv) Delta Learning Rule

(CO1)

Q2. a. Explain the architecture and algorithm of Single Layer Perceptron. Discuss its limitations with suitable examples. (CO2)

b. Describe Feedback Networks with detailed explanation of Hopfield Network and Bidirectional Associative Memory (BAM). (CO2)

c. Explain Feedforward Networks including Backpropagation Network (BPN) and Radial Basis Function Network (RBFN). Also discuss Self-Organizing Maps (SOM) and Learning Vector Quantization (LVQ). (CO2)

Q3. a. Define Fuzzy Sets and explain their properties and operations with suitable examples. (CO3)

b. Explain Fuzzy Relations, including cardinality, types, and properties with mathematical representation. (CO3)

c. Describe Fuzzy Composition and explain its significance in fuzzy systems with examples. (CO3)

Q4. a. Define Fuzzy Variables and explain different types of Membership Functions with diagrams. (CO4)

b. Explain Fuzzy Rule-Based Systems with emphasis on Mamdani and Takagi-Sugeno models. (CO4)

c. Describe the Fuzzy Inference System (FIS) process including:

- (i) Fuzzification
- (ii) Rule Base
- (iii) Inference Engine
- (iv) Defuzzification

(CO4)



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- Q5. a. Explain Biological Terminology used in Genetic Algorithms and relate it to computational concepts. (CO5)
- b. Describe the elements of Genetic Algorithm, including encoding, selection methods, crossover types, mutation, and reinsertion strategies. (CO5)
- c. Explain the working of a Simple Genetic Algorithm and discuss its theoretical foundation:
- (i) Schema Theorem
 - (ii) Fundamental Theorem of GA
 - (iii) Building Block Hypothesis (CO5)